Sine, cosine, and tangent ratios

1 $\angle ACB$

2

a $\frac{24}{7}$

b $\frac{12}{5}$

3 $\frac{4}{5}$

4

a $\frac{5}{13}$

b $\frac{12}{13}$

5

a $\frac{7}{25}$

b $\frac{7}{25}$

6

a $\frac{4}{5}$

b $\frac{4}{5}$

7

a $21 : 23$

b $21 : 23$

c $23 : 31.14$

d $63 : 93.43$

8 $\frac{\text{opposite}}{\text{hypotenuse}}$

9

a $\sin\theta = \frac{AC}{AB}$

b $\cos\theta = \frac{BC}{AB}$

c $\tan\theta = \frac{AC}{BC}$

10

a $\tan\theta = \frac{8}{15}$

b $\cos\theta = \frac{15}{17}$

11

a $\tan\theta = \frac{3}{4}$

b $\sin\theta = \frac{4}{5}$

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12

a $\cos\theta = \frac{5}{13}$

b $\sin\theta = \frac{24}{25}$

c $\tan\theta = \frac{7}{4}$

d $\tan\theta = \frac{24}{7}$

15

a $\angle BCA$

b $\cos\theta = \frac{3}{5}$

c $\tan\theta = \frac{4}{3}$

16

a $\angle BCA$

b $x = 8$

c $\sin\theta = \frac{4}{5}$

d $\tan\theta = \frac{4}{3}$

17

a $x = 13$

b $\sin\theta = \frac{12}{13}$

c $\cos\theta = \frac{5}{13}$

18

a

i $x = 4$

ii $\tan\theta = \frac{4}{3}$

b

i $x = 12$

ii $\tan\theta = \frac{5}{12}$

19

a Yes

b Yes

c No

20 $\frac{3.5}{25} \leq \tan x \leq \frac{5}{25}$

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Additional questions

21

a
i BC ii AC iii BA

b
i X ii Y iii Z

c
i Y ii Z iii X

d
i b ii a iii c

22

a $\angle C$

b $\angle B$

c $\angle B$

d $\angle A$

23

a AC

b BC

c BC

d AC

e $\angle ACB$

24

a
i $\frac{21}{29}$ ii $\frac{20}{29}$ iii $\frac{21}{20}$

b
i $\frac{12}{13}$ ii $\frac{5}{13}$ iii $\frac{12}{5}$

c
i $\frac{5}{7}$ ii $\frac{4.9}{7}$ iii $\frac{5}{4.9}$

d
i $\frac{4}{5}$ ii $\frac{3}{5}$ iii $\frac{4}{3}$

e
i $\frac{9}{31}$ ii $\frac{29.7}{31}$ iii $\frac{9}{29.7}$

f
i $\frac{21}{29}$ ii $\frac{20}{29}$ iii $\frac{21}{20}$

25

a $\angle BCA$

b $\frac{8}{17}$

c $\frac{15}{17}$

26 From the diagram we have the following relationships: $\sin\theta = \frac{12}{13}, \cos\theta = \frac{5}{13}$ $\tan\theta = \frac{12}{5}$.

$\frac{\sin\theta}{\cos\theta} = \frac{\frac{12}{13}}{\frac{5}{13}} = \frac{12}{13} \times \frac{13}{5} = \frac{12}{5}$ which is equal to $\tan\theta$ as required.

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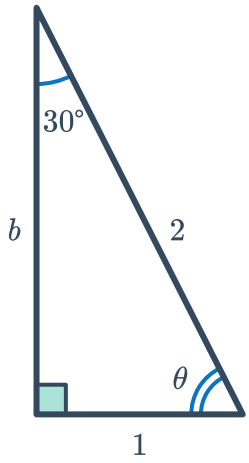
27 a $\dfrac{BC}{AB}$ b $\dfrac{AC}{AB}$

c $\dfrac{BC}{AC}$

28 We can see that $\sin\theta = \dfrac{3}{5}$ and $\cos\alpha = \dfrac{3}{5}$.
This means $\sin\theta = \cos\alpha$. This is because the opposite side in reference to θ is the adjacent side in reference to α and the hypotenuse must be the same.

29 The two triangles are similar triangles as enlargement is a similarity transformation. This means that their side lengths are all in the same ratio.
As an example, let's look at the right triangle with side lengths $3, 4, 5$. If $\cos\theta = \dfrac{3}{5}$ then the cosine ratio of the corresponding angle would be equal to $\dfrac{12}{20} = \dfrac{3}{5}$ which is the same value as $\cos\theta$ in the smaller triangle. We can prove the sine and tangent ratios are equal in a similar manner.

30



Any right triangle that has an angle measure of 30° must have an angle with a measure of 60°. This means it is a 30°-60°-90° triangle, and the ratio of sides in a 30°-60°-90° triangle is $1 : \sqrt{3} : 2$. The side opposite the angle with measure of 30 must be the shortest side of the triangle, and the adjacent side must be the middle length. This means $\tan 30^\circ = \dfrac{1}{\sqrt{3}}$, so by rationalizing the denominator $\tan 30^\circ = \dfrac{\sqrt{3}}{3}$

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